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Ensemble deep learning models for stock price forecasting: evidence from the Saudi markets

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ABSTRACT

Investments are the foundation of financial markets, directing resources toward activities that generate future returns. Stock trading plays a central role in wealth creation, yet forecasting stock prices remains difficult due to the nonlinear and volatile nature of financial data. This study presents an ensemble deep learning framework for short-term stock price forecasting in the Saudi market. It addresses the limited use of advanced ensemble methods in emerging economies. The framework uses multivariate data from ten companies listed on the Tadawul All Share Index and compares two ensemble strategies: averaging and stacking with standalone models (Gated Recurrent Unit, Long Short-Term Memory, and Bidirectional Recurrent Neural Network). A total of 34 engineered technical indicators and a rolling-window setup were applied for five-day-ahead predictions. The results show that ensemble models consistently outperform individual architectures across all evaluation metrics. Stacking ensembles, especially those combining Gated Recurrent Unit and Long Short-Term Memory, achieved the highest accuracy. Averaging ensembles provided stable and efficient alternatives. Overall, the findings confirm the robustness and adaptability of ensemble methods for financial forecasting and offer practical insights for investors and policymakers seeking reliable, data-driven prediction tools.

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INTRODUCTION

Economic forecasting emerged in the early 20th century as a tool for studying business cycles. Today, it remains vital for governments seeking to anticipate indicators such as industrial production, unemployment, and inflation. Accurate forecasts enable policymakers to design informed fiscal and monetary strategies. In the private sector, businesses use forecasting to plan production, manage inventories, and attract investors by improving financial performance and revenue stability. Investments are the foundation of financial markets, channeling resources toward activities that generate future returns. Among available instruments, stocks are the most common and represent ownership in companies. However, forecasting stock prices is inherently complex because markets are

volatile, nonlinear, and influenced by numerous external factors. Over time, investors have adopted two main analytical approaches: technical analysis, which relies on historical prices and market patterns, and fundamental analysis, which evaluates economic and political factors to estimate intrinsic value.

Advances in computational methods have significantly improved forecasting accuracy. Traditional statistical models such as ARIMA and GARCH remain widely used for time-series prediction but are constrained by their linear assumptions, making them less effective for complex and noisy financial data. In contrast, machine learning (ML) and deep learning (DL) techniques can model nonlinear dependencies and adapt to evolving market dynamics. Models such as Artificial Neural Networks (ANNs), Gated Recurrent Units (GRUs), and Long Short-Term Memory (LSTM) networks have shown strong capabilities in capturing temporal patterns in stock price movements. However, the application of ensemble deep learning (EDL), which integrates multiple DL models to enhance robustness and accuracy, remains relatively new and underexplored in the financial forecasting domain.

Most prior research focuses on univariate forecasting, using only historical prices. However, multivariate approaches that include additional technical indicators (TIs) or macroeconomic variables tend to achieve better predictive performance and robustness. Studies have shown that combining these indicators helps capture hidden market dependencies and short-term volatility patterns that univariate models often overlook. As noted by *Kumbure et al. (2022)*, most existing studies center on the U.S. market, although interest in Asian markets such as China, Japan, and India is increasing. Despite this progress, empirical research on Arabian Gulf markets, particularly Saudi Arabia, remains limited, especially in multivariate using EDL. The Saudi capital market, one of the world's fastest-growing investment hubs (*Narayanan, 2023*), provides a unique opportunity to explore the effectiveness of ensemble forecasting in an emerging economy.

This study aims to evaluate the performance of EDL models for short-term stock price forecasting. Building on our previous work (*Baamer & Aljamaan, 2025*), which examined multivariate forecasting using standalone DL models with TIs, this research extends the framework by integrating ensemble techniques, specifically averaging and stacking, to improve robustness and predictive accuracy. By focusing on the Saudi market, the study contributes to the growing literature on applying advanced ML to emerging financial systems.

The remainder of this article is structured as follows: 'Background' and 'Literature Review' presents theoretical background and review related work. 'Research Methodology' and 'Empirical Study' outline the research design and ensemble framework. 'Results and Discussion' presents results and discussion. 'Threats to Validity' addresses study limitations, and finally 'Conclusion' concludes with key findings and future directions.

BACKGROUND

This section outlines the study background, summarizing key concepts in time series forecasting, stock market prediction, and EDL methods supporting the proposed framework.

Time series forecasting

Time series forecasting is a key technique in data science and is widely used in business, finance, supply chain management, manufacturing, and inventory planning. A time series represents a collection of data points recorded at successive time intervals, such as seconds, minutes, hours, days, or months, arranged in chronological order. The main goal of time series analysis is to understand the underlying dynamics and detect trends, patterns, and anomalies to forecast future values.

Two main approaches exist for time series forecasting: univariate and multivariate. Univariate forecasting uses one variable to predict its future values, whereas multivariate forecasting involves multiple interrelated variables influenced by both time and each other. Multivariate approaches capture complex dependencies and generally improve predictive accuracy in real-world scenarios. Forecasting one time step ahead is referred to as one-step forecasting, while predicting multiple future values is called multi-step forecasting. Although more challenging due to error accumulation, multi-step forecasting better reflects real financial and economic dynamics.

Stock price forecasting

Stock is an essential investment type representing a unit in a company's ownership. Therefore, forecasting stock prices is necessary for the financial field. For decades, investors have been attempting to apply various strategies and techniques for boosting profits. These movements were investigated in terms of two groups during the analysis.

First, technical analysis can be defined as forecasting targetable price changes based on previous price changes. This analysis is employed in markets with high volatility, like the Foreign Exchange Market (FOREX). Technical Indicator considers indicators such as closing, open, high, and low prices as well as volume and applies them to equations to construct other technical variables such as returns, stochastic oscillators, moving averages, and the relative strength index (RSI). Second, the fundamental analysis approach forecasts the anticipated price movements of financial data using statistical data from the analysis and other aspects, such as the economy, politics, the environment, and other factors.

Ensemble learning

Ensemble learning (*Zhou, 2025*) combines multiple models to produce predictions that are more accurate and robust than those of any individual model, as shown in Fig. 1. Ensembles are commonly divided into two types: competitive and cooperative. Competitive ensembles train independent models separately and then aggregate their predictions, often through mechanisms such as averaging or voting. Cooperative ensembles divide the prediction task into sub-tasks handled by specialized models, and the outputs of these specialized models are later combined. Selecting suitable base models, determining the number of models, and choosing an appropriate integration strategy are critical design decisions in ensemble learning.

Ensemble methods are generally categorized into basic and advanced techniques. Basic ensemble methods include max voting, averaging, and weighted averaging. Max voting is mainly used for classification tasks, where each model votes for a class and the majority

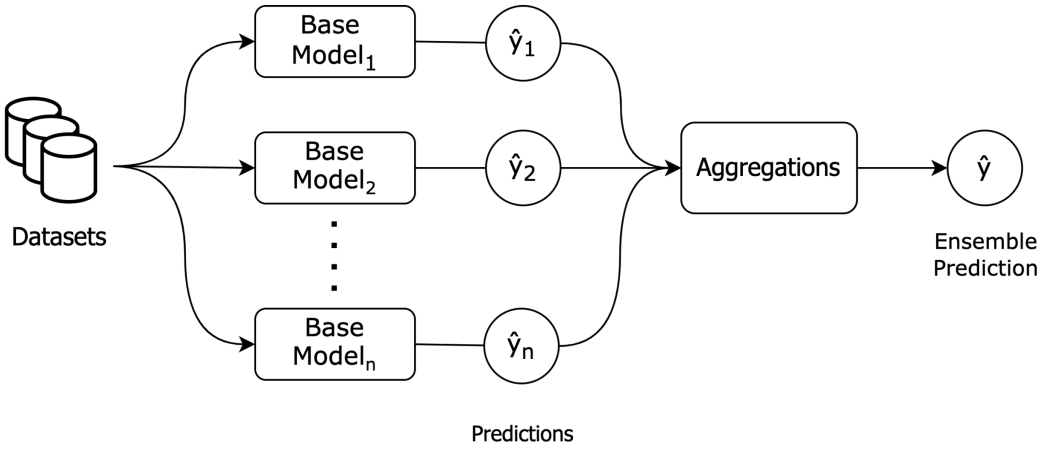


Figure 1 **Illustration of an ensemble.** Full-size DOI: 10.7717/peerj-cs.4064/fig-1

decision is selected as the final output. Averaging is commonly used for regression, where the final prediction is the mean of all model outputs. Weighted averaging extends simple averaging by assigning each model a weight that reflects its performance, so that stronger models contribute more to the final prediction.

Advanced ensemble methods rely on more sophisticated integration strategies to enhance performance. In stacking (STK), the predictions of multiple base models are combined and passed to a meta-learner, which generates the final output. Blending is conceptually similar to stacking, but it uses a validation subset of the training data instead of the full training or test data when fitting the meta-learner. Bagging trains multiple versions of a model on bootstrapped subsets of the data to reduce variance; Random Forests are a well-known example of this approach. Boosting trains models sequentially, where each new model focuses on correcting the errors made by the previous one. Common boosting implementations include AdaBoost, Gradient Boosting Machines, XGBoost, LightGBM, and CatBoost.

LITERATURE REVIEW

Over the past three decades, numerous studies have examined stock price forecasting. However, predicting financial time series remains challenging because such data are non-stationary, nonlinear, and influenced by external market dynamics. Research in this domain can be grouped by its objective, either price forecasting (a regression problem) or movement prediction (a classification problem). The current study focuses on price forecasting, which involves predicting continuous stock values rather than their directional trends.

Standalone machine learning models

ANNs have been widely used for stock forecasting and often outperform classical methods such as ARIMA and Support Vector Machine (SVM) (*Singh & Srivastava, 2017*). Despite this, ANNs face challenges in modeling noisy, high-dimensional financial data, sometimes leading to overfitting and unstable results (*Shen et al., 2011*; *Larochelle et al., 2009*).

Their shallow architectures with limited hidden layers restrict their ability to capture deeper temporal dependencies. Recent studies have shifted toward DL architectures, which provide greater flexibility and representation power. *Sezer, Gudelek & Ozbayoglu (2020)* conducted a systematic review (2005–2019) highlighting Recurrent Neural Networks (RNNs), especially LSTM models, as dominant in time series forecasting. Similarly, *Kumbure et al. (2022)* reported a surge in DL applications from 2005–2021, with models such as stacked LSTM and GRU showing strong performance (*Althelaya, El-Alfy & Mohammed, 2018*).

Machine learning ensembles

Ensemble learning has long been proposed as a strategy to improve model robustness and generalization (*Chun & Park, 2005*). Early approaches such as Dynamic Adaptive Ensemble Case-Based Reasoning (DAE-CBR) optimized parameter combinations for better forecasts. Other studies compared homogeneous ensembles (*e.g.*, neural networks) and heterogeneous ensembles (*e.g.*, logistic regression, decision trees, SVMs) using techniques like bagging and voting (*Tsai et al., 2011*). *Barak, Arjmand & Ortobelli (2017)* applied multi-classifier ensemble frameworks that combined Boosting, Bagging, and AdaBoost, improving classification and regression accuracy. Ensemble tree-based methods such as Random Forest and XGBoost were also found effective in predicting both price direction and intraday returns (*Khaidem, Saha & Dey, 2016*; *Basak et al., 2019*; *Lohrmann & Luukka, 2019*).

Further research expanded ensemble integration. *Nti, Adekoya & Weyori (2020a)* compared 25 ensemble configurations, including stacking, blending, bagging, and boosting across multiple stock exchanges, showing that stacking and blending achieved superior accuracy. Similarly, *Nti, Adekoya & Weyori (2020b)* proposed a GA-SVM hybrid that used genetic algorithms for feature selection and SVM optimization, achieving improved forecasts on the Ghana stock market. Subsequent studies refined feature engineering within ensemble systems (*Ampomah, Qin & Nyame, 2020*; *Yun, Yoon & Won, 2021*; *Lin et al., 2021*), confirming that careful feature design plays a central role in achieving superior accuracy.

Deep learning ensembles

Recent research trends integrate DL models into ensemble architectures for stock forecasting. For instance, *Shah et al. (2020)* combined LSTM and GRU models to merge technical and fundamental features, improving short-term prediction accuracy. *Althelaya, Mohammed & El-Alfy (2021)* introduced a two-stage stacked LSTM, where multiple temporal resolutions were fused to enhance performance. Similarly, *Li & Pan (2022)* designed a blended ensemble of recurrent networks followed by a dense layer, outperforming traditional bagging. Weighted averaging and hybrid approaches have also shown promise. *Kumar & Anitha (2022)* integrated LSTM, BiLSTM, and FB Prophet using weighted averaging, achieving strong results on Google stock data. Comprehensive review by *Kontopoulou et al. (2023)* identified LSTM and GRU as state-of-the-art for modeling long-term dependencies in multivariate financial data.

Saudi market forecasting

Research on Saudi stock forecasting remains relatively limited but is steadily increasing. Early works relied on univariate models, whereas recent studies explore multivariate frameworks that include multiple indicators. *Jarrah & Derbali (2023)* demonstrated that multivariate LSTM models significantly outperform univariate ones for seven-day forecasts. Likewise, *Mohammed (2022)* found that poorly chosen TIs can harm model performance, underscoring the importance of effective feature selection. *Aseeri (2023)* developed a GRU-based forecasting model with a structured feature engineering pipeline and compared it to VARMAX. Their analysis highlighted persistent methodological limitations in prior work, such as small sample sizes (*Jarrah & Salim, 2019*), improper evaluation metrics (*Alotaibi et al., 2018*), and weak data transparency (*Alamro, McCarren & Al-Rasheed, 2019*). Further, *Yuvaraj et al. (2025)* integrated RNNs with short-term moving averages (EMA_5 and EMA_10), achieving notable accuracy across MAE, RMSE, and R^2 metrics.

Building on these findings, our previous study (*Baamer & Aljamaan, 2025*) proposed a comprehensive DL framework for multivariate forecasting on Tadawul data. The models: GRU, LSTM, and BiRNN were tested with 34 TIs, and results showed GRU and LSTM outperforming both BiRNN and traditional baselines. We also introduced an ensemble averaging (EA) technique that achieved the lowest error rates across all companies, surpassing previous benchmarks such as *Aseeri (2023)*. A comparative summary of major Saudi forecasting studies is presented in Table 1, highlighting datasets, features, model types, and top-performing algorithms. The collective evidence indicates that ensemble DL frameworks, especially those combining GRU and LSTM, offer the most promising direction for robust and generalizable stock price forecasting in the Saudi market.

Research gaps

Despite extensive research on stock price forecasting, several key gaps remain across existing methodologies. First, most prior studies on ML and DL models rely on standalone architectures, such as ANNs, LSTMs, or GRUs. While these models effectively capture temporal dependencies, they often struggle with overfitting and instability when applied to noisy, high-dimensional financial data. The absence of model integration has resulted in inconsistent predictive performance across different markets and time horizons. Second, traditional ensemble approaches, including bagging, boosting, and blending, have shown improvements in accuracy and generalization but largely depend on shallow learners and manual feature engineering. These limitations restrict their scalability and reduce their ability to model complex, nonlinear dependencies typical of financial time series.

Third, recent progress in DL ensembles has demonstrated strong potential; however, most applications remain focused on mature markets such as the U.S. and China. Few studies have explored multivariate EDL frameworks that integrate TIs and recurrent architectures within emerging markets. Finally, within the Saudi market, empirical studies remain limited and often rely on small datasets, narrow feature sets, or single-model architectures. EDL models combining multiple recurrent structures have not yet been systematically evaluated for short-term, multivariate stock forecasting. Addressing these

Table 1 Literature comparison of Saudi stock forecasting studies.

Study	Target companies	Period	Features	Predicted value	Models	Performance measurements	Outperforming model
Baamer & Aljamaan (2025)	Table 3	2001–05/2020	34 TIs	5-day ahead	BiRNN, GRU, GRU (256 units) (*Aseeri, 2023*), LSTM, EA	MAE, MSE, RMSE, R^2	EA
Aseeri (2023)	Table 3	2001–05/2020	14 features 9 HD & 5 TIs: WMA, DEMA, TEMA, HMA, ZLEMA	5-day ahead	VARMAX, GRU	MAE, MSE, RMSE, R^2	GRU
Jarrah & Derbali (2023)	STC, Mobily, Zain	01/2017–11/2022	4 HD	7-day ahead	U-LSTM, M-LSTM	MAE, MAPE, MSE, RMSE	M-LSTM
Mohammed (2022)	SISCO, Almarai, Al Rajhi	01/2010–08/2022	5 TIs: MA, MACD, RSI, STO, BB	1-day ahead	B-LSTM, GRU, S-LSTM	MAE, MAPE, MSE, R^2	S-LSTM
Al-Nefaie & Aldhyani (2022)	Communication, Energy, Financial, Industrial.	01/2018–12/2020	7 HD	1-day ahead	MLP, LSTM	MSE, RMSE, R^2	LSTM
Jarrah & Salim (2019)	SAMC, Sabic, Riyad Bank, Yanbu Cement, Rajhi Bank, Saudi Indian	01/2011–03/2016	5 HD	7-day ahead	(DRNN), ARIMA	MAE, MSE, RMSE	DRNN

Note:
Abbreviations: HD, Historical data; TIs, technical indicators; EA, ensemble averaging; DRNN, deep recurrent neural network; U-LSTM, univariate LSTM; M-LSTM, multivariate LSTM; B-LSTM, bidirectional LSTM; S-LSTM, stacked LSTM.

gaps, this study develops and evaluates a comprehensive EDL framework using averaging and stacking strategies to improve forecasting robustness and accuracy in the Saudi stock market.

RESEARCH METHODOLOGY

This study addresses the challenge of multi-step multivariate stock price forecasting by evaluating the performance of various EDL models in predicting stock prices. The focus is on forecasting accuracy, using regression-based metrics, from the perspectives of investors and researchers. The analysis involves stock prices from ten selected companies across different sectors in the Saudi stock market (Tadawul).

Research questions

To achieve our research objective, two key research questions are formulated.

- **RQ 1.** To what extent can EDL techniques outperform standalone DL models to improve the accuracy and reliability of stock forecasting in the Saudi market?
  Rationale. This RQ investigates the effectiveness of EDL techniques compared to standalone DL models in improving accuracy and reliability for stock forecasting within the Saudi market. It aims to determine whether the complexity and resources needed for the ensemble methods lead to significantly enhanced forecast performance, given the unique characteristics of the Saudi market.

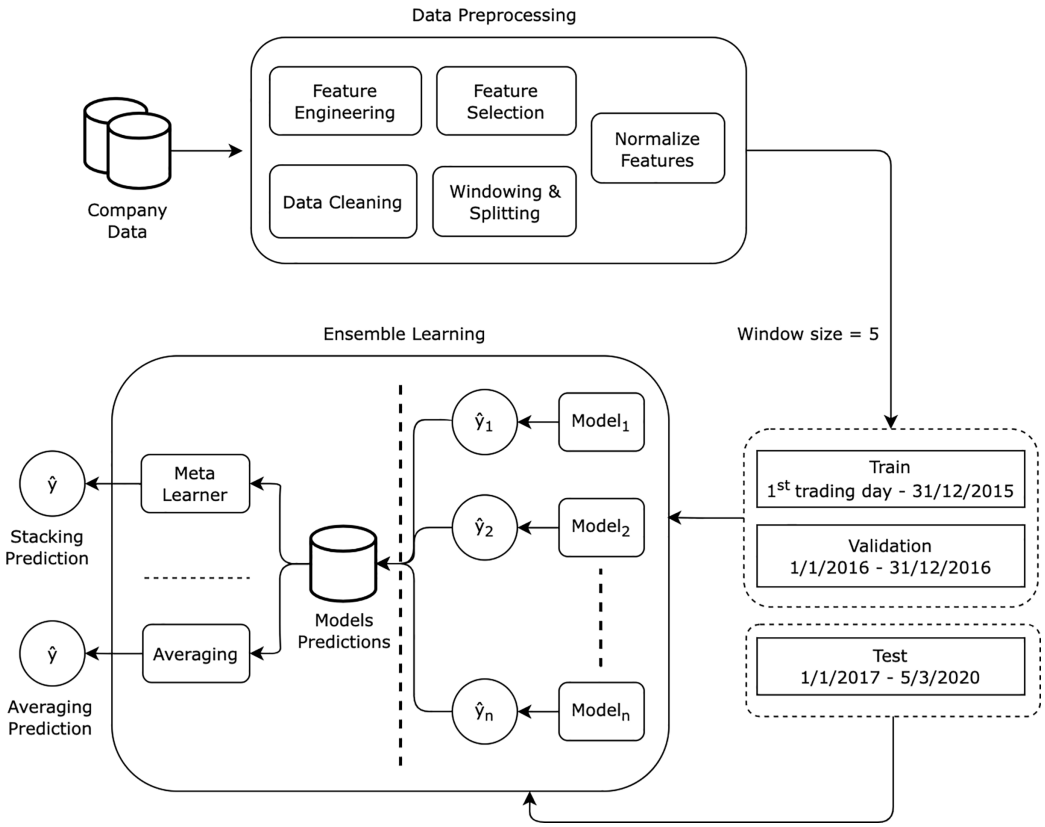


Figure 2 Overview of the proposed forecasting framework. Historical stock data are first processed through a preprocessing pipeline consisting of data cleaning (handling missing values with the fill-forward technique), feature engineering using technical indicators, feature selection based on correlation analysis, normalization using Standard scaling, and windowing with an input window size of $w = 5$ trading days. The dataset is then chronologically split into training, validation, and testing periods to preserve temporal order. Multiple deep learning base models (BiRNN, GRU, and LSTM) generate predictions $\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n$, which are combined using two ensemble strategies: simple averaging and stacking, where a meta-learner integrates the base model outputs to produce the final forecast $\hat{y}$. Full-size DOI: 10.7717/peerj-cs.4064/fig-2

- **RQ 2.** To what extent can the proposed EDL models, developed using Saudi market data, be effectively applied to a different financial market environment, such as the U.S. stock market?

  Rationale. This RQ evaluates the robustness of EDL architectures across distinct market environments. It investigates whether the models trained and optimized using Saudi market data maintain competitive performance when applied to selected U.S. stocks. This question examines the external applicability of the ensemble framework and its potential suitability for broader financial forecasting contexts.

Methodology steps

Figure 2 illustrates our study methodology steps. The methodology starts with data preprocessing pipeline, where company data undergo feature engineering and selection, missing data imputation, and normalization. Then, the dataset is then divided into

Table 2 Description of the TASI dataset.

Sector	Company name	Trading name	Start date	End date	# Records
Industrials	Al-Ahsa Development Co.	ADC	31/12/2001	05/03/2020	4,756
Financials	Al Rajhi Bank	ALRAJHI	31/12/2001	05/03/2020	4,758
Real estate	Emaar The Economic City	EMAAR EC	07/10/2006	05/03/2020	3,346
Consumer discretionary	Fitaihi Holding Group	FITAIHI GROUP	26/01/2002	05/03/2020	4,420
Consumer staples	Al Gassim Investment Holding Co.	GACO	01/01/2002	05/03/2020	4,738
Energy	Saudi Arabia Refineries Co.	SARCO	23/01/2002	05/03/2020	4,458
Utilities	Saudi Electricity Co.	SAUDI ELECTRICITY	08/06/2002	05/03/2020	4,627
Healthcare	Saudi Pharmaceutical Industries	SPIMACO	31/12/2001	05/03/2020	4,758
Communication services	Saudi Telecom Co.	STC	25/01/2003	05/03/2020	4,433
Materials	National Industrialization Co.	TASNEE	31/12/2001	05/03/2020	4,759

training, validation, and testing sets. In the feature engineering process, a large number of TIs are computed, and then reduced to 34 key indicators. Then, dataset is structured into windows of size 5 (*Baamer & Aljamaan, 2025*). In the ensemble learning phase, we train a number of base models to generate independent predictions (*Baamer & Aljamaan, 2025*). These outputs are then combined using two ensemble strategies: simple averaging and stacking, where the latter refines forecasts through additional meta-learners. The specific details of our methodology are presented next in the empirical study section.

EMPIRICAL STUDY

This section presents our empirical study design on multi-step multivariate stock price forecasting within the context of the Saudi stock market. This study aims to examine how EDL will enhance prediction performance compared to standalone DL models. A replication package containing all code and experimental artifacts used in this study is publicly available at our GitHub repository. For long-term preservation and version control, the repository has been archived on Zenodo with DOI: https://doi.org/10.5281/zenodo.19046571.

Dataset description

Effectively utilizing extensive and well-processed historical data (HD) from time series is important for achieving optimal forecasting results, especially in time-series analysis. In this study, we employed the multivariate time series dataset of the Tadawul All Share Index (TASI), the primary index of the Saudi stock market which is publicly available *via* *Alzahrani (2020)*. The dataset is widely used for stock price forecasting and provides comprehensive market insights. As shown in Table 2, ten companies were selected from each sector, representing Saudi enterprises across ten industries (*Baamer & Aljamaan, 2025*). The dataset includes more than 13 features which are described in Table 3.

Data preprocessing

As shown in Fig. 2, the initial preprocessing steps followed those in *Baamer & Aljamaan (2025)* study. A comprehensive feature engineering process was conducted to generate over

Table 3 TASI dataset features.

Feature	Type	Description
Symbol	Integer	The identifier or reference number of the company.
Name	String	The full name of the company.
Trading_name	String	The trading (short) name of the company.
Sector	String	The sector in which the company operates.
Date	Date	The date corresponding to the stock price record.
Open	Numeric	The opening stock price of the day.
High	Numeric	The highest stock price reached during the day.
Low	Numeric	The lowest stock price reached during the day.
Close	Numeric	The closing stock price of the day.
Change	Numeric	The absolute change in price compared to the previous trading day.
Perc_change	Numeric	The percentage change in price compared to the previous trading day.
Volume_traded	Numeric	The total volume of shares traded during the day.
Value_traded	Numeric	The total monetary value of trades during the day.
No_trades	Numeric	The total number of individual trades executed during the day.

130 TIs provided by the `pandas-ta` Python library (*Johnson, 2018*). These indicators include a wide range of indicators derived from TA-Lib and candlestick pattern analytics.

To reduce redundancy and enhance model performance, feature selection was conducted using Pearson's correlation coefficient (*Hall, 1999*). Correlation coefficients were computed exclusively on the training data for each company. Validation and test partitions were strictly excluded from this process to prevent information leakage. Only indicators exhibiting a strong linear correlation ($r > 0.8$ or $r < -0.8$) with the target variable (closing price) across all ten companies were retained. This threshold ensures the inclusion of features demonstrating consistent and statistically meaningful relationships with price behavior, thereby improving generalizability while reducing noise. Pearson correlation was employed as an initial feature filtering mechanism to eliminate weakly associated or redundant indicators and reduce dimensionality.

While Pearson measures linear associations, it was not intended to capture nonlinear dependencies. Instead, nonlinear interactions among the retained indicators are subsequently modeled by the deep learning architectures (GRU, LSTM, and BiRNN), which are inherently capable of learning complex nonlinear relationships. Thus, correlation-based selection serves as a noise-reduction and dimensionality control step rather than a limitation on model expressiveness. The final set of 34 selected technical indicators is presented in Table 4 for transparency and reproducibility.

Feature selection was applied using a global correlation threshold consistently across all companies. While company-specific selection could potentially capture firm-level predictive signals, our objective was to construct a unified and generalizable forecasting framework applicable across multiple stocks. Using a consistent feature subset enhances cross-company comparability, reduces overfitting risk, and supports robust ensemble evaluation across markets.

Table 4 List of the 34 selected technical indicators used in the Saudi market experiments.

TI	Name	Description
open	_	The price at the start of the trading session.
high	_	Highest price in a trading session.
low	_	Lowest amount in a trading session.
close	_	The closing price of the previous trading session.
alma	_	Moving average uses the curve of Normal(Gauss) distribution.
decay	_	A process send signals from the past into the future.
dema	Double Exponential Moving Average	Produce an average that is smoother and has less latency.
ema	Exponential Moving Average	Exhibits greater responsiveness.
fwma	Fibonacci's Weighted Moving Average	The weights are based on the Fibonacci Sequence.
hma	The Hull Exponential Moving Average	Attempts to remove lag in moving averages.
hl2	_	The average of the highest and lowest prices for a period.
hlc3	_	Average of High, Low, and Close for a bar/candle.
hwma	Holt-Winter Moving Average	A three-parameter moving average by the Holt-Winter method.
jma	Mark Jurik's Moving Average	Attempts to eliminate noise to see the "true" underlying activity.
kama	Kaufman's Adaptive Moving Average	A moving average designed to account for market noise or volatility.
linreg	Linear Regression Moving Average	A simplified version of a Standard Linear Regression.
median	_	Rolling Median of over 'n' periods.
midpoint	_	Halfway between the day's highest high and lowest low.
midprice	_	Mid price = (Highest High + Lowest Low)/2
pwma	Pascal's Weighted Moving Average	Similar to a symmetric triangular window.
quantile	_	Divides a dataset into intervals containing equal portions.
rma	The WildeR's Moving Average	It is an Exponential Moving Average (EMA) modifying alpha
sinwma	_	A weighted average using sine cycles.
sma	The Simple Moving Average	It is the equally weighted average over n periods.
ssf	Ehler's Super Smoother Filter	Two-pole super smoother filter
swma	Symmetric Weighted Moving Average	Weights are based on a symmetric triangle
t3	Tim Tillson's T3 Moving Average	It is considered a smoother and more responsive moving average.
tema	Triple Exponential Moving Average	A less laggy Exponential Moving Average.
trima	Triangular Moving Average	The shape of the weights are triangular
vwap	The Volume Weighted Average Price	That measures the average typical price by volume.
vwma	Volume Weighted Moving Average	SMA that assigns more weight to periods of higher volume.
wcp	Weighted Closing Price	The weighted price given: high, low and double the close.
wma	The Weighted Moving Average	The weights are linearly increasing.
zlma	Zero Lag Moving Average	Attempts to eliminate the lag associated with moving averages.

The selected features were then standardized to remove the mean and scale the data to unit variance, ensuring that all variables contributed equally during model training. To preserve temporal causality, standardization was conducted using a train-only fitting strategy. The scaler was fitted on the training set of each company and subsequently applied to the validation and test sets using the same parameters. At no point were statistics from future observations used during preprocessing.

Table 5 Average RMSE across companies for different historical window sizes.

Window size (w)	Average RMSE
3	0.047
5	0.043
7	0.051
10	0.044

The forecasting task was formulated as a multivariate regression problem, targeting the prediction of closing prices over a forecast horizon of five trading days ($h = 5$), corresponding to a full business week. To evaluate the impact of the historical window size, a sensitivity analysis was conducted using alternative window sizes ($w = 3, 5, 7, 10$). As shown in Table 5, a historical window size of $w = 5$ achieved the lowest average RMSE across companies. Although the difference between $w = 5$ (0.043) and $w = 10$ (0.044) is marginal, $w = 5$ was selected because it provides slightly better predictive performance while naturally aligning with the five-day forecasting horizon (one trading week) adopted in this study. Larger windows increase input complexity without meaningful forecasting gains. Therefore, $w = 5$ was retained in the final experimental configuration. Predictions are based on a historical time-lag window of size w, capturing relevant prior observations across all features. A one-stride sliding window approach was applied iteratively across the multivariate time series (*Aseeri, 2023*; *Baamer & Aljamaan, 2025*). This process is defined as:

$$y_{t+h} = f_h(y_t, y_{t-1}, \ldots, y_{t-w}).$$

For clarity and consistency, Table 6 summarizes the mathematical symbols used in the forecasting formulation.

The forecasting task was implemented using a direct multi-output strategy. Specifically, each model produces a vector of five future closing prices $(t + 1, \ldots, t + 5)$ in a single forward pass. This design enables the model to learn dependencies across future time steps simultaneously and avoids the error accumulation associated with recursive forecasting approaches. No separate models were trained per horizon. All preprocessing steps, including feature selection and normalization, were performed independently for each company to avoid cross-company information leakage.

EDL models

The experimental environment was developed using Python 3. For our experiments, we utilized DL models created with the Keras library, which operates on TensorFlow version 2.15. Based on earlier experiments documented in *Baamer & Aljamaan (2025)*, we determined the top-performing BiRNN (denoted as B), GRU (denoted as G), and LSTM (denoted as L) for each company included in the study. The selection of GRU, LSTM, and BiRNN was guided by their established effectiveness in financial time-series forecasting and their ability to model sequential dependencies with moderate computational complexity. While alternative architectures such as Temporal Convolutional Neural Network or Transformers are promising, they typically require larger datasets and

Table 6 Notation used in the forecasting and baseline formulations.

Symbol	Description
t	Current time index
y_t	Observed closing price at time t
y_{t+h}	Closing price at future time $t + h$
w	Historical window size (number of past observations used as input)
h	Forecast horizon (number of steps ahead)
$f_h(\cdot)$	Forecasting function for horizon h
δ	Constant drift term in the Random Walk model
ε_t	White noise error term at time t
p	Autoregressive order in the ARIMA model
d	Differencing order used to achieve stationarity in ARIMA
q	Moving average order in the ARIMA model
L	Lag operator in ARIMA formulations
$\Phi_p(L)$	Autoregressive polynomial of order p
$\Theta_q(L)$	Moving average polynomial of order q
x_t	Feature vector at time t used by machine learning models
$\hat{y}_t$	Predicted value of the target variable at time t
K	Number of trees in the XGBoost ensemble
$f_k(\cdot)$	The k-th regression tree in the XGBoost model

introduce higher architectural complexity. Given the study's focus on ensemble strategy evaluation rather than exhaustive architecture benchmarking, recurrent models were selected to ensure methodological consistency and controlled experimental design. This study aimed to explore EDL by utilizing the best-performing models as foundational components. We applied averaging and stacking techniques, elaborately discussed in the background section, to enhance model performance and robustness.

Averaging ensemble

We constructed our averaging ensembles by combining different permutations of the three DL models (B, G, and L). This study investigates which average ensemble provides the best prediction performance across companies. Specifically, we examined all possible pairwise and triplet combinations of these models, resulting in four averaging ensembles per company (*Baamer & Aljamaan, 2025*): AVG(B,G,L), AVG(B,G), AVG(B,L), and AVG(G,L). For each model combination, we computed the average of the forecasts generated by the corresponding DL models. In total, four averaging ensembles were developed for each company, resulting in 40 models used for stock price forecasting under the averaging ensemble approach. The performance of these ensembles was then evaluated against the actual test data using the selected evaluation metrics.

Stacking ensemble

Stacking is an ensemble learning approach that combines predictions from multiple base models to train a meta-learner, which generates the final predictions on the test data. In this study, we built the stacking using three DL models (B, G, and L) as base models, and

utilized three types of meta-learners: ANN, GRU, and LSTM, to train on the base models predictions. Base model selection for stacking was conducted using validation performance within the training phase only, without exposure to the test set, thereby preventing selection bias. The rationale for selecting top-performing models was to reduce redundancy and avoid introducing weak or highly correlated predictors into the meta-learner, which may degrade ensemble stability. Preliminary analysis showed that including all base models did not yield systematic improvements and occasionally increased variance; therefore, a filtered subset was retained for the final stacking configuration. The ANN meta-learner comprised three fully connected layers with 128, 64, and 32 units, respectively, each employing the ReLU activation function. The GRU and LSTM meta-learners each contained a single recurrent layer with 128 units. All meta-learners were trained using the Adam optimizer and MSE loss function to ensure consistent optimization and fair comparison across architectures.

To prevent target leakage in the stacking procedure, we adopted a rolling-origin out-of-fold (OOF) protocol tailored for time-series data. Specifically, base models were first trained using an expanding window strategy on the training partition. For each fold, the model was trained on past observations and generated predictions on the immediately following validation block. These out-of-fold validation predictions were collected across all folds and used as meta-features to train the meta-learner. Importantly, no test data were used during base-model training or meta-learner fitting. During the final evaluation phase, base models were retrained on the full training data and applied to the unseen test set. Their predictions were then fed into the trained meta-learner to produce the final stacked forecasts. This rolling-origin stacking protocol ensures strict temporal ordering and prevents information leakage across training, validation, and test partitions.

Table 7 shows the twelve stacking ensemble constructed for each company. For evaluation, the test set predictions were combined according to these configurations, and the trained meta-learners were applied to generate the final forecasts. The choice of ANN, GRU, and LSTM as meta-learners reflects the multivariate and multistep nature of the forecasting problem, where each base model produces multiple five-day-ahead predictions. This design allows the stacking ensemble to capture both temporal dependencies and cross-model interactions to enhance the forecast accuracy and robustness.

Training configuration

To ensure full reproducibility and methodological transparency, we explicitly report the training configuration used across all deep learning models. The importance of carefully selecting and reporting training hyperparameters has been highlighted in prior stock forecasting studies, as these settings can substantially influence predictive performance and model generalization (*Hoque & Aljamaan, 2021*). Table 8 summarizes the hyperparameters and optimization settings applied consistently throughout the experiments.

The model corresponding to the lowest validation loss was selected for final evaluation on the unseen test set.

Table 7 Stacking ensembles with different meta-learners and base model combinations.

Stacking model	Base models	Meta-learner
STK_ANN(B,G,L)	B, G, L	ANN
STK_ANN(B,G)	B, G	ANN
STK_ANN(B,L)	B, L	ANN
STK_ANN(G,L)	G, L	ANN
STK_GRU(B,G,L)	B, G, L	GRU
STK_GRU(B,G)	B, G	GRU
STK_GRU(B,L)	B, L	GRU
STK_GRU(G,L)	G, L	GRU
STK_LSTM(B,G,L)	B, G, L	LSTM
STK_LSTM(B,G)	B, G	LSTM
STK_LSTM(B,L)	B, L	LSTM
STK_LSTM(G,L)	G, L	LSTM

Table 8 Training configuration for deep learning models.

Component	Setting
Optimizer	Adam
Learning rate	1×10^{-4}
Loss function	MSE
Batch size	10% of the train set
Epochs	100
Early stopping	20
Activation function	ReLU

Baseline models

To ensure a rigorous evaluation of the proposed forecasting framework, three baseline models were considered: the Random Walk, the AutoRegressive Integrated Moving Average (ARIMA), and Extreme Gradient Boosting (XGBoost). These baselines represent three distinct modeling paradigms commonly used in financial forecasting: naive stochastic benchmarks, classical statistical time-series models, and modern machine learning approaches.

Random Walk Model: In financial time series forecasting, the Random Walk model serves as a fundamental baseline due to its simplicity and strong theoretical grounding in the Efficient Market Hypothesis. The Random Walk model assumes that stock prices fully incorporate all available information at each time step, implying that future price movements cannot be systematically predicted from historical observations. Consequently, price changes follow a stochastic process where successive increments are independent.

The Random Walk with drift can be expressed as

$$y_t = y_{t-1} + \delta + \varepsilon_t,$$

where y_t denotes the stock price at time t, δ represents a constant drift term capturing the average directional movement of prices, and ε_t is a random error term reflecting market noise, both defined in Table 6, where all symbols used in the formulation are defined.

In this study, the drift component was fixed to align with the five-day forecasting horizon, providing a transparent and interpretable benchmark consistent with common Random Walk formulations in financial forecasting literature. Although adaptive drift estimation based on rolling volatility could be considered, the baseline was intentionally kept parsimonious to maintain a clear point of comparison with the proposed EDL models. Despite its simplicity, the Random Walk model often performs competitively in short-term financial forecasting and therefore provides an important reference point for assessing the practical utility of more complex models.

ARIMA Model: To provide a classical statistical benchmark, the AutoRegressive Integrated Moving Average (ARIMA) model was also employed. ARIMA models are widely used in financial time series analysis because of their ability to capture linear temporal dependencies, autocorrelation structures, and stochastic trends in non-stationary data.

The general ARIMA(p, d, q) model can be written as

$$\Phi_p(L)(1 - L)^d y_t = \Theta_q(L)\varepsilon_t,$$

where L denotes the lag operator, p represents the autoregressive order, d is the order of differencing required to achieve stationarity, and q denotes the moving average order. The autoregressive component models the dependence of the current value on previous observations, while the moving average component accounts for correlations between the current error term and past forecast errors.

ARIMA models are particularly effective in modeling financial time series exhibiting persistence and short-term autocorrelation. In this study, ARIMA was used as a purely univariate statistical benchmark based solely on historical price observations. This allows a direct comparison between traditional econometric forecasting techniques and the proposed deep learning ensemble framework.

XGBoost Model: In addition to statistical baselines, Extreme Gradient Boosting (XGBoost) was included as a representative tree-based machine learning benchmark. XGBoost is a scalable implementation of gradient boosted decision trees that constructs an ensemble of weak learners sequentially, where each new tree attempts to minimize the residual errors produced by previous trees.

The model prediction can be expressed as

$$\hat{y}_t = \sum_{k=1}^{K} f_k(x_t),$$

where f_k denotes the k-th regression tree and x_t represents the feature vector at time t. Model training is performed by minimizing a regularized objective function composed of a loss component and a complexity penalty that controls model overfitting.

XGBoost has demonstrated strong performance across a wide range of tabular prediction tasks due to its ability to capture nonlinear relationships and complex feature

Table 9 Evaluation metrics used to assess model performance.

Evaluation metric	Equation
Mean absolute error (MAE)	$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\lvert y_i - \hat{y}_i\rvert$
Mean squared error (MSE)	$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$
Root mean squared error (RMSE)	$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$
Mean relative error (MRE)	$\text{MRE} = \frac{1}{N}\sum_{i=1}^{N}\left\lvert\frac{y_i - \hat{y}_i}{y_i}\right\rvert$

interactions. In this study, XGBoost utilizes the same multivariate feature set employed by the deep learning models, including historical price variables and technical indicators. This provides a competitive machine learning benchmark against which the performance gains of the proposed EDL models can be evaluated.

Evaluation measures

Stock price forecasting is treated as a regression problem and evaluated using five standard metrics: MAE, MSE, RMSE, and MRE. MAE measures the average absolute deviation between predicted and actual prices, while MSE squares these errors to penalize larger deviations. RMSE, the square root of MSE, maintains the original data scale and is more interpretable for financial applications. MRE expresses the average relative error as a proportion of the true value, enabling fair comparison across stocks with different price ranges. The mathematical definitions of these metrics are summarized in Table 9.

RESULTS AND DISCUSSION

This section presents and discusses the empirical findings of five-day-ahead forecasting for ten TASI-listed companies. Two ensemble strategies averaging and stacking were evaluated against standalone deep learning models (GRU, LSTM, and BiRNN). To provide contextual benchmarks, two reference models were included: a Random Walk with five-day drift serving as a naive baseline, and a GRU (256 units) benchmark model (*Aseeri, 2023*).

Model performance was assessed using multiple evaluation metrics, with particular emphasis on RMSE due to its sensitivity to larger errors and its relevance in financial risk contexts. As this study is framed as a regression-based multistep forecasting task rather than a trading strategy design, evaluation focuses on continuous error measures (RMSE, MAE, and MRE), which are standard in financial time-series literature. Since RMSE is expressed in the same units as the price series, it enables direct monetary interpretation of forecast deviations. Consequently, reductions in RMSE correspond to lower average pricing errors, which are economically meaningful in short-term forecasting settings where misestimation may influence risk exposure and decision-support processes. Although trading simulations are beyond the scope of this study, improved RMSE reflects enhanced forecast reliability. From an economic perspective, RMSE values represent the average

Table 10 Top performing models per company across evaluation metrics.

Company	Model (RMSE)	RMSE	Model (MAE)	MAE	Model (MRE)	MRE
ADC	STK_GRU(B,G)	0.2498	STK_GRU(B,G)	0.1585	STK_GRU(B,G)	0.0135
ALRAJHI	STK_GRU(B,G)	1.0688	STK_GRU(B,G)	0.7269	STK_GRU(B,G)	0.0131
EMAAR EC	AVG(B,G)	0.3259	AVG(B,G)	0.2154	AVG(B,G)	0.0179
FITAIHI GROUP	AVG(G,L)	0.1908	AVG(G,L)	0.1252	AVG(G,L)	0.0105
GACO	STK_LSTM(B,L)	0.2637	AVG(B,G)	0.1727	AVG(B,G)	0.0144
SARCO	AVG(B,G)	1.2307	AVG(B,G,L)	0.7587	AVG(B,G)	0.0194
SAUDI ELECTRICITY	STK_GRU(B,G)	0.4651	STK_GRU(B,G)	0.3046	STK_GRU(B,G)	0.0150
SPIMACO	AVG(B,G)	0.5192	AVG(B,G)	0.3484	AVG(B,G)	0.0115
STC	STK_LSTM(B,G)	1.9293	STK_LSTM(B,G)	1.2476	STK_ANN(B,G)	0.0142
TASNEE	STK_LSTM(B,G)	0.4401	STK_GRU(B,G,L)	0.3068	STK_LSTM(B,G)	0.0189

magnitude of price deviation in the same monetary units as the underlying stock prices. Therefore, improvements in RMSE directly translate into more accurate short-term price estimates, which may support risk management, portfolio monitoring, and decision-support applications. While this study does not explicitly simulate trading strategies, lower forecasting errors indicate more reliable price projections that could improve downstream financial decision processes.

Across all companies, ensemble models consistently outperformed both baseline and reference models. The baseline Random Walk with Drift model produced the highest forecasting errors across most datasets, confirming its limited suitability for multistep stock price prediction. As a persistence-based model, Random Walk assumes that future prices follow recent observations with a constant drift component, which prevents it from capturing nonlinear dynamics and structural patterns commonly observed in financial time series.

Classical statistical and machine learning baselines also demonstrated weaker predictive performance relative to deep learning approaches. Although ARIMA models are effective for capturing linear temporal dependencies, their reliance on linear assumptions restricts their ability to model nonlinear relationships and interactions among technical indicators. Similarly, XGBoost achieved stronger predictive performance than Random Walk and ARIMA by leveraging gradient boosting to capture nonlinear feature interactions. However, its performance remained generally inferior to recurrent neural network models, which are specifically designed to learn temporal dependencies in sequential financial data.

The GRU(256) benchmark was also surpassed by nearly all deep learning alternatives, suggesting that individual architectures benefit significantly from architectural diversity and ensemble integration. In particular, stacking-based ensembles, especially *STK_GRU(B,G)* and *STK_LSTM(B,G)*, achieved the lowest RMSE values (Table 10), demonstrating their ability to effectively capture complex temporal patterns. For example, *STK_GRU(B,G)* achieved the best performance for ADC, ALRAJHI, and SAUDI ELECTRICITY, while averaging ensembles delivered strong and stable results across EMAAR EC, SPIMACO, FITAIHI GROUP, and SARCO.

Comparisons across RMSE, MAE, and MRE (Table 10) revealed highly consistent ranking patterns, with only minor variations observed for GACO, SARCO, and TASNEE. Even in these cases, the best-performing models still belonged to the same ensemble families either stacking or averaging highlighting the robustness and stability of ensemble learning across multiple evaluation metrics and datasets.

The superior performance of ensemble models can be partially attributed to variance reduction, as combining diverse base learners stabilizes predictions and mitigates model-specific noise. However, the consistent advantage of stacking over simple averaging indicates that the improvements cannot be explained solely by variance reduction. Instead, the meta-learner in stacking adaptively integrates heterogeneous temporal representations produced by different base models, enabling more effective exploitation of complementary predictive information.

The particularly strong performance of GRU-based ensembles may be theoretically explained by the architectural characteristics of GRU networks. Compared with LSTM, GRU employs fewer gating mechanisms and parameters, which can reduce the risk of overfitting when training on moderate-sized financial datasets. Since daily stock price series typically exhibit short- to medium-term temporal dependencies, the simplified memory structure of GRU can efficiently capture relevant temporal dynamics without introducing unnecessary model complexity. Within ensemble frameworks, this architectural efficiency may lead to more stable base learners and improved overall ensemble performance.

Overall, the experimental results demonstrate that EDL, particularly stacking architectures combining GRU and LSTM representations, significantly improves short-term stock price forecasting accuracy in the Saudi stock market. While averaging ensembles provide a computationally simpler and robust alternative, stacking ensembles consistently deliver the highest predictive accuracy across most datasets.

Model ranking *via* Wilcoxon test

To evaluate the statistical robustness of the forecasting models, pairwise comparisons were conducted using the Wilcoxon signed-rank test. This non-parametric test is appropriate for financial forecasting evaluation because prediction errors in financial time series often violate the normality assumptions required by parametric statistical tests. The Wilcoxon test has also been widely used in forecasting studies to compare predictive models when the distribution of forecasting errors is unknown or non-Gaussian.

Forecasting performance was summarized using RMSE computed over fixed five-step forecasting windows. Although the forecasting models are trained using a one-stride sliding window during the learning phase, the statistical comparison relies on non-overlapping evaluation windows when computing RMSE sequences. This design was applied consistently throughout the statistical evaluation to better satisfy the independence assumptions required for the Wilcoxon signed-rank test. Using window-level RMSE provides a suitable unit of comparison for multistep forecasting tasks, where individual prediction errors within the same horizon may be temporally correlated. The use of

Table 11 Aggregated win-loss statistics derived from Wilcoxon signed-rank tests across all companies and forecasting horizons after Holm–Bonferroni correction for multiple comparisons.

Model	Wins	Losses	Win rate (%)
AVG(B,G)	4,058	520	88.64
AVG(B,G,L)	3,947	520	88.36
STK_GRU(B,G)	3,864	544	87.66
STK_GRU(B,G,L)	3,661	556	86.82
AVG(B,L)	3,646	609	85.69
GRU	3,408	717	82.62
STK_LSTM(B,G,L)	3,473	743	82.38
AVG(G,L)	3,466	790	81.44
STK_GRU(B,L)	3,429	789	81.29
STK_LSTM(B,L)	3,300	804	80.41
BRNN	3,325	983	77.18
LSTM	3,157	1,875	62.74
XGBoost	6,191	3,067	66.87
Random Walk	7,659	17,252	30.75
Benchmark	2,065	16,227	11.29
ARIMA	430	27,930	1.52

non-overlapping windows further reduces temporal dependence among observations and better approximates the independence assumptions required for statistical testing.

The resulting RMSE sequences were used to perform pairwise comparisons across the evaluated models for each company and forecasting horizon. Because a large number of pairwise comparisons were conducted, Holm–Bonferroni correction was applied to control the family-wise error rate associated with multiple testing. Statistical significance was assessed at a corrected significance level of $\alpha = 0.05$.

Although non-overlapping forecasting windows were used throughout the statistical evaluation to better satisfy the independence assumptions of the Wilcoxon signed-rank test, some residual dependence may still exist because all forecasts originate from the same underlying financial time series. Nevertheless, the adopted evaluation protocol provides a more appropriate basis for statistical comparison than overlapping forecast windows. To provide an interpretable comparison across a large number of models, the outcomes of the pairwise Wilcoxon tests were summarized using a win–loss framework. In this analysis, each statistically significant comparison contributes one win to the model with the lower RMSE and one loss to the competing model. Aggregating these outcomes across companies and forecasting horizons provides a robust indication of the relative consistency of model performance.

In addition to statistical significance, the magnitude of performance differences can also be interpreted from the aggregated win–loss statistics, which summarize consistent superiority across multiple datasets and forecasting horizons. Table 11 summarizes the aggregated win–loss statistics across all companies and forecasting horizons. Models with higher win counts indicate stronger and more consistent predictive performance across

datasets. The results show that ensemble-based models dominate the ranking. In particular, the averaging ensembles AVG(B,G) and AVG(B,G,L) achieved the highest win rates (88.64% and 88.36%, respectively), followed closely by GRU-based stacking ensembles such as STK_GRU(B,G) (87.66%) and STK_GRU(B,G,L) (86.82%). These findings indicate that combining multiple deep learning architectures through ensemble learning consistently improves predictive robustness.

Several additional ensemble variants, including AVG(B,L) and AVG(G,L), also achieved win rates above 80%, further reinforcing the effectiveness of ensemble aggregation strategies. Among the standalone deep learning models, GRU achieved the strongest performance (82.62%), outperforming both LSTM (62.74%) and BiRNN (77.18%). Traditional baselines showed substantially weaker statistical performance. The Random Walk model achieved a win rate of 30.75%, while the benchmark model obtained only 11.29%. ARIMA exhibited the lowest statistical performance with a win rate of 1.52%, indicating limited competitiveness compared to modern machine learning and deep learning approaches. Although XGBoost records a relatively large number of wins due to the high number of pairwise comparisons, its overall win rate remains substantially lower than ensemble models and GRU-based architectures, indicating weaker overall consistency across forecasting tasks.

Overall, the Wilcoxon analysis corroborates the RMSE-based evaluation results: ensemble architectures particularly averaging ensembles and GRU-based stacking models consistently deliver superior and statistically robust forecasting performance across companies and forecast horizons. These results provide consistent statistical evidence that ensemble-based deep learning architectures offer improved robustness and predictive stability compared with standalone models and traditional baselines.

RQ1 answer: Ensemble deep learning models achieved the strongest and most consistent forecasting performance across companies. Averaging ensembles and GRU-based stacking models consistently ranked among the top performers, outperforming standalone deep learning models as well as traditional baselines such as Random Walk, benchmark models, and ARIMA.

Computational cost analysis

In addition to predictive performance, we analyzed the computational cost of the evaluated models by measuring both training time and inference time across all companies. Table 12 reports the average computational cost for each model. The results indicate that traditional statistical models such as Random Walk with Drift and ARIMA require minimal training time compared to deep learning models. However, ARIMA shows relatively higher inference time due to its iterative forecasting mechanism.

Deep learning models exhibit significantly higher training costs. In particular, the LSTM model required the longest training time (approximately 488 s on average), followed by BRNN and GRU models. This increase in computational cost is expected due to the

Table 12 End-to-end average training and inference time of the evaluated models across all companies. For ensemble models, the reported time includes both the cost of the constituent base models and the additional ensemble stage.

Model	Total training time (s)	Total inference time (s/sample)
Random walk with drift	0.00	0.021
ARIMA	1.06	2.99
XGBoost	17.65	0.030
BRNN	185.66	1.186
GRU	175.54	0.993
LSTM	488.76	2.524
AVG(B,G)	361.20	2.179
AVG(B,L)	674.42	3.710
AVG(G,L)	664.30	3.517
AVG(B,G,L)	849.96	4.703
STK_ANN(B,G,L)	858.84	4.937
STK_ANN(B,G)	370.27	2.764
STK_ANN(B,L)	683.11	3.934
STK_ANN(G,L)	673.25	3.758
STK_GRU(B,G,L)	866.71	5.584
STK_GRU(B,G)	377.89	2.758
STK_GRU(B,L)	691.06	4.253
STK_GRU(G,L)	681.37	4.101
STK_LSTM(B,G,L)	866.72	5.320
STK_LSTM(B,G)	377.95	2.746
STK_LSTM(B,L)	691.01	4.218
STK_LSTM(G,L)	680.87	4.069

complexity of recurrent neural network architectures. Regarding ensemble approaches, stacking ensembles introduce additional training overhead because of the meta-learner training stage. Depending on the architecture of the meta-model, stacking required approximately 8 to 17 s of additional training time. In contrast, the averaging ensemble introduces virtually no additional training overhead, as it simply aggregates predictions from pre-trained base models. Despite the additional training cost, the inference time of stacking ensembles remains relatively small, making both ensemble strategies suitable for practical stock price forecasting applications.

Generalization results

To assess the generalizability of the proposed forecasting framework beyond the Saudi stock market, the experimental pipeline was replicated on selected U.S. companies using the publicly available S&P 500 dataset (*Mooney, 2022*), a widely recognized benchmark of U.S. market performance. The dataset includes standard financial variables such as Date, Open, High, Low, Close, Adjusted Close, and Volume. Four representative companies were selected Apple (AAPL), Bank of America (BAC), Alphabet Inc. (GOOG), and Johnson &

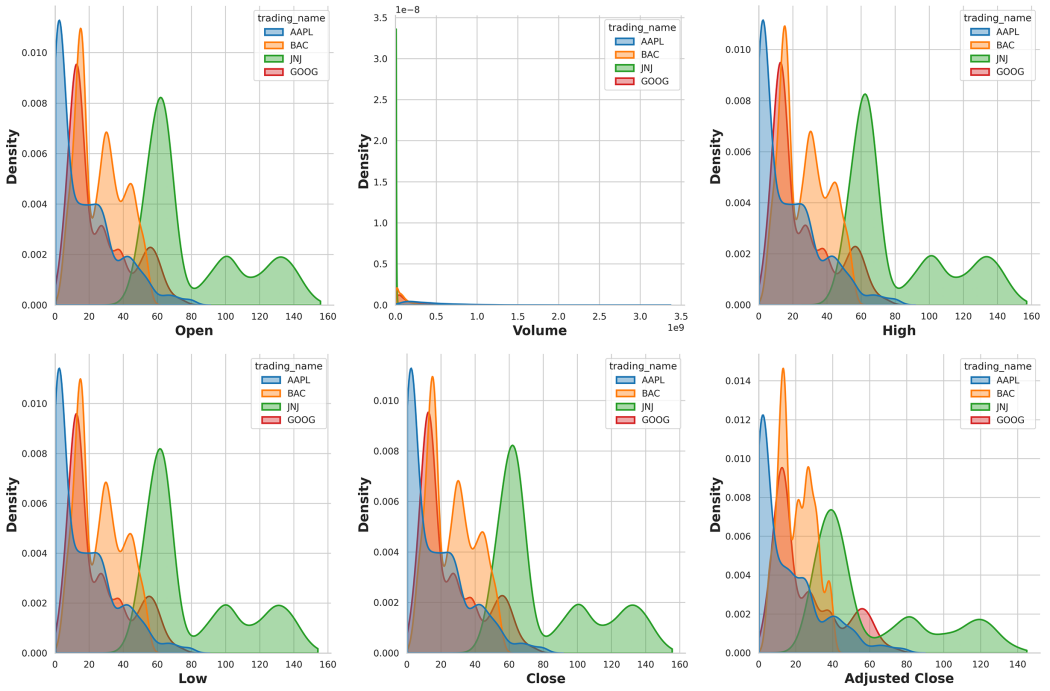


Figure 3 Distribution of key financial features (open, volume, high, low, close, and adjusted close) for the selected U.S. companies (AAPL, BAC, GOOG, and JNJ). Kernel density estimation (KDE) curves are used to visualize the empirical distribution of each feature across companies. The plots illustrate the variability and heterogeneity of the U.S. market data used in the forecasting experiments.
Full-size DOI: 10.7717/peerj-cs.4064/fig-3

Johnson (JNJ) with the time period aligned to the Saudi dataset, covering December 31, 2001 to March 5, 2020.

The same experimental protocol used in the Saudi market experiments was retained. The data were partitioned into training data up to December 31, 2015, validation data from January 1, 2016 to December 31, 2016, and testing data from January 1, 2017 to March 3, 2020. The modeling pipeline illustrated in Fig. 2 was applied consistently, including preprocessing, feature engineering, model architectures, and evaluation metrics. The feature engineering process generated a set of technical indicators comparable to those used in the Saudi experiments. While most indicators overlapped across datasets, some market-specific variables differed due to data availability and structural characteristics of the markets. For example, indicators such as JMA, KAMA, and SSF appeared only in the Saudi dataset, whereas variables such as Adjusted Close and Positive Volume Index (PVI) were unique to the U.S. dataset.

It is important to note that the selected U.S. companies do not constitute a statistically representative sample of the entire U.S. market. The purpose of this experiment is therefore not to establish broad statistical generalization across the U.S. market, but rather to provide an external validation of the proposed forecasting framework under a different financial environment. As illustrated in Fig. 3, the U.S. stocks exhibit higher price variability compared to their Saudi counterparts, which may increase forecasting difficulty.

Table 13 Top performing models across US companies based on RMSE.

Company	Model	MAE	MRE	RMSE
AAPL	STK_ANN(B,G,L)	0.8243	0.0172	1.2127
BAC	GRU	0.4802	0.0171	0.6954
GOOG	STK_ANN(B,L)	0.8156	0.0148	1.1786
JNJ	AVG(B,G)	1.4857	0.0111	2.1984

Performance metrics across AAPL, BAC, GOOG, and JNJ varied accordingly. AAPL and BAC were relatively more predictable, showing lower forecasting errors across evaluation metrics, whereas GOOG and particularly JNJ exhibited higher prediction errors, suggesting more complex price dynamics and higher volatility.

Table 13 presents the best-performing models for each U.S. company based on RMSE. Stacking ensembles using ANN architectures achieved the highest accuracy for AAPL and GOOG, while the standalone GRU model performed best for BAC. For JNJ, the averaging ensemble AVG(B,G) produced the lowest RMSE. These findings indicate that although ensemble models generally perform well, the optimal configuration may vary depending on company-specific characteristics. Overall, ensemble models particularly simple averaging and GRU-based stacking architectures demonstrated consistently competitive performance across both the Saudi and U.S. datasets. Although some variation in model rankings was observed, likely driven by differences in market volatility and company-specific behaviors, the strong performance of these ensemble approaches across both environments supports the robustness of the proposed forecasting framework.

RQ2 answer: Ensemble deep learning models maintained strong predictive performance across the examined U.S. companies. While the best-performing configuration varied by company, averaging ensembles and GRU-based stacking consistently ranked among the top-performing models, supporting the cross-market applicability of the proposed forecasting framework.

Scope and Future Extensions: This study focuses on price-level forecasting formulated as a multistep regression problem. Forecasting returns or log-returns represents a related but statistically different task, typically characterized by higher volatility and weaker autocorrelation structures. While ensemble-based improvements are expected to remain beneficial due to variance reduction properties, quantitative performance metrics may differ under return-based formulations.

Financial markets are also subject to structural breaks and regime shifts, particularly during extreme events such as the COVID-19 crisis or major commodity shocks. Although crisis-specific sub-periods were not isolated in this study, the dataset includes episodes of elevated volatility. Ensemble methods may provide enhanced stability under such turbulent conditions due to aggregation effects, though their performance during extreme

structural breaks warrants dedicated investigation. Finally, the proposed ensemble framework can be extended to online or incremental learning environments. Base learners may be periodically updated using sliding or expanding windows, while stacking meta-learners can be retrained as new observations become available. Although this study evaluates an offline setting for controlled comparison, online adaptation represents a promising direction for future financial forecasting applications.

THREATS TO VALIDITY

This study, while offering valuable insights into stock price forecasting in the Saudi market, faces several threats to validity. By acknowledging these threats and adopting multiple ensemble strategies and evaluation metrics, this study aims to enhance the transparency and replicability of its findings. External validity may be limited due to market-specific characteristics such as volatility, liquidity, and trading behavior. However, replication of the experiments on U.S. stock data yielded promising generalizability, supporting the robustness of the proposed models across different financial environments. Internal validity may be influenced by the reliance on a specific historical period, which might not fully capture broader macroeconomic cycles or regime shifts. Additionally, the forecasting models were trained on TIs derived from historical price movements, which may inherently carry temporal biases.

Construct validity is challenged by the use of Pearson correlation for feature selection, which primarily detects linear relationships and may overlook important nonlinear interactions between features and stock price behavior. Despite this limitation, the ensemble models, especially averaging and stacking approaches, help mitigate this risk by combining the strengths of multiple base learners, potentially capturing complex dependencies more effectively. Conclusion validity is generally supported by consistent model performance across multiple error metrics, namely RMSE, MAE, and MRE. While minor variations were observed, all three metrics yielded comparable evaluation results and led to consistent conclusions regarding model rankings and forecasting accuracy. In addition, statistical comparisons were conducted using non-overlapping evaluation windows and corrected for multiple hypothesis testing to mitigate risks of inflated statistical significance. Nevertheless, risks such as overfitting remain possible, although they were addressed through techniques like dropout regularization and cross-validation.

CONCLUSION

This study investigated the effectiveness of EDL models for multistep stock price forecasting in the Saudi stock market using multivariate TASI data. Two ensemble strategies averaging and stacking were evaluated against standalone deep learning architectures (GRU, LSTM, and BiRNN) as well as traditional baselines including Random Walk, ARIMA, and XGBoost. Forecasting performance was assessed across ten Saudi companies using multiple evaluation metrics, with particular emphasis on RMSE due to its economic interpretability in financial forecasting contexts.

The empirical results demonstrate that ensemble approaches consistently outperform standalone models and traditional baselines across most datasets. In particular, averaging

ensembles such as AVG(B,G) and AVG(B,G,L) achieved the highest overall win rates in the statistical ranking analysis, while GRU-based stacking ensembles also exhibited strong predictive performance. The Wilcoxon signed-rank tests confirmed the statistical robustness of these findings, indicating that ensemble architectures provide more stable and reliable forecasts across companies and forecasting horizons. Among the standalone models, GRU achieved the strongest predictive performance, outperforming both LSTM and BiRNN. Traditional benchmarks showed substantially weaker results. While XGBoost improved over linear statistical models by capturing nonlinear feature interactions, its performance remained generally inferior to recurrent neural network models. Similarly, ARIMA and Random Walk exhibited limited forecasting capability in the multistep setting, highlighting the advantages of deep learning approaches for modeling complex temporal dependencies in financial data.

The computational cost analysis revealed an expected trade-off between predictive accuracy and computational efficiency. Stacking ensembles achieved high predictive accuracy but required higher training and inference costs due to the additional meta-learning stage. In contrast, averaging ensembles provided a computationally efficient alternative while still delivering highly competitive forecasting performance. This suggests that averaging ensembles may represent a practical compromise between accuracy and efficiency in real-world forecasting applications. To evaluate the generalizability of the proposed framework, the experimental pipeline was replicated on selected U.S. companies from the S&P 500 dataset. The results showed that ensemble models maintained strong predictive performance across different market environments, although the optimal model configuration varied by company. These findings indicate that ensemble-based deep learning architectures can provide robust forecasting performance across diverse financial markets. Overall, the results demonstrate that EDL constitutes an effective strategy for improving short-term stock price forecasting accuracy. By combining heterogeneous temporal representations learned by different recurrent architectures, ensemble models reduce prediction variance and enhance forecast stability. These advantages are particularly valuable in financial markets where data are noisy and nonlinear dynamics are prevalent.

Future research may extend this work by investigating additional ensemble strategies, including bagging and boosting approaches, as well as alternative meta-learners for stacking architectures. Expanding the analysis to a larger set of financial assets and exploring return-based forecasting formulations may also provide further insights into the robustness and applicability of EDL methods in financial time-series forecasting.

MATERIALS AND METHODS

Experiments were conducted using Google Colaboratory (Colab), a cloud-based Jupyter notebook environment. The computational setup is summarized as follows: Platform: Google Colaboratory, Operating System: Linux (Ubuntu-based, managed by Colab), CPU: Intel Xeon (cloud virtualized, exact model varies by session), RAM: Approximately 12–16 GB (standard Colab runtime), GPU: NVIDIA Tesla T4/P100 (when GPU runtime is

enabled), Python Version: Python 3.9, Deep Learning Framework: TensorFlow/Keras, Execution Environment: Jupyter Notebook (cloud-based).

ADDITIONAL INFORMATION AND DECLARATIONS

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Competing Interests

The authors declare that they have no competing interests.

Author Contributions

- Rana Baamer conceived and designed the experiments, performed the experiments, analyzed the data, performed the computation work, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.
- Hamoud Aljamaan conceived and designed the experiments, analyzed the data, prepared figures and/or tables, authored or reviewed drafts of the article, and approved the final draft.

Data Availability

The following information was supplied regarding data availability:

The code is available at GitHub and Zenodo:

- https://github.com/ranabameer/saudi-stock-forecasting-ensembles/

- Baamer, R. (2026). Saudi Stock Price Forecasting Using Deep Learning Ensembles (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19046571

The data is available at Kaggle: https://www.kaggle.com/datasets/salwaalzahrani/saudi-stock-exchange-tadawul.

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